WAYHUNTER

புதிய முழுமையான வணிகத் தேவைகள் ஆவணம்

MASTER BUSINESS REQUIREMENTS DOCUMENT • TAMIL EDITION

USER • FRIENDS • TRAVELER • HUNTER • GAME • COINS • ADMIN • APIs

மூல ஆவணங்கள்: Wayhunter_Final_BRD.pdf (18 பக்கங்கள்); WayHunter Admin Portal Full Module Specification (20 பக்கங்கள்); Wayhunter API Integration Groups Tamil.docx (12 குழுக்கள்).

மாற்றங்களுக்கான அடிப்படை: 07 நவம்பர் 2026 அன்று வழங்கப்பட்ட புதிய Product Owner Requirements. இந்த ஆவணம் Development மற்றும் Supervisor Review-க்கான ஒருங்கிணைந்த Draft; முழுமையான இறுதி அங்கீகாரம் இன்னும் பெறப்படவில்லை.

Technical stack: Flutter Mobile | Angular Admin | ASP.NET Core Web API + SignalR | SQL Server/EF Core | Cloudinary | Google/Firebase & other approved integrations

VERSION 2.1 • CONTROLLED DRAFT UPDATE • 10 NOV 2026

ஆவணக் கட்டுப்பாடு மற்றும் முக்கிய விளக்கம்

அம்சம்	புதிய நிலை
Document name	Wayhunter Master BRD - Revised Tamil v2.1 (voucher change-controlled draft)
Primary market	Sri Lanka
Owner	Wayhunter Project Team / Product Owner
Status	Draft - revised requirements; owner approval / development feasibility / QA / supervisor sign-off pending
Scope	Flutter User App + Angular Admin Portal + 12 API/Integration Groups
Precedence	புதிய User Decisions > 05 Nov Mobile BRD > Admin Specification > Implementation recommendations. முரண்பாடுகள் decision log-ல் பதிவு.
Currency	Coins only - HP, RC, XP, Lifetime HP UI/business logic பழையதாகக் கருதப்படும்
Security baseline	Server-side decisions, consent, role checks, immutable or append-only audit / ledger

இந்த ஆவணத்தை எப்படிப் படிக்க வேண்டும்?

"USER DECISION" என்பது இப்போது தெளிவாகக் கூறப்பட்ட புதிய விதி; "INHERITED" என்பது மாற்றமில்லாமல் எடுத்துள்ள பழைய BRD விதி; "RECOMMENDED" என்பது implementation பாதுகாப்புக்கான முன்மொழிவு; "DECISION REQUIRED" என்பது இன்னும் இறுதி தீர்மானம் தேவைப்படும் அம்சம். இந்த வேறுபாட்டைத் தெளிவாக வைத்திருக்க வேண்டும்.

புதிய மாற்றங்களின் சுருக்கம்

Code	Change	Impacted Mobile / Admin
CH-01	ஒரே Friends Page: Profile + Trip Add Friends; Followers/Following; invite share; opted-in contacts discovery	P-07, T-02/T-12, User Admin, notifications
CH-02	Hunter: live capture only, My Hunts statuses, every new submission to admin with user ID + name	H-02..H-13, Admin 04, 06, 11
CH-03	Game: admin publish/finish approvals, My Games status+history, one-month visibility	G-01..G-16, Admin 10, 11
CH-04	Leaderboard 3 sources, global shortcut disabled in specified pages	Traveler/Hunter/Game, navigation
CH-05	Coins-only economy, levels at 2000 earned Coins, rewards and vouchers	Profile, W-01..W-08, Admin 02/11/12/13/17
CH-06	Registration-issued unique user ID; all approval requests labeled ID+name	Auth, Admin queues and audits
CH-07	Friend notifications; route hazard pop-up; trip-start reminders; actual start/end validation	Notifications, Traveler, Admin 08/09/16
CH-08	Include full 17 admin modules and 12 source-listed API groups	Admin & Integration appendix
CH-09	Voucher image/PDF source verification clarified; OCR/QR/barcode only proposed; debit timing conflict flagged VCH-DEC-01.	ADM-12/13, Wallet, Backend, Angular Admin, ERD appendix

உள்ளடக்க அட்டவணை

01 User Identity, Details & Authentication
02 Common Home and Navigation
03 Friends & Social Graph
04 Traveler: 14 Existing Views + New Leaderboard
05 Location Hunter: 14 Views
06 Game: 16 Views
07 Leaderboards
08 Profile & Settings: 18 Views
09 Coins Wallet, Redemption & Vouchers
10 Notifications & Hazard Alerts
11 Admin Portal: 17 Modules and UJs
12 Cross-module Business Rules
13 State Transitions and Data Model
14 12 API Integration Groups
15 Security, Reliability, Cost & KPI
16 Acceptance Tests
17 Conflicts, Pending Decisions and Change Traceability
18 Sources and Sign-off
19 Controlled Voucher Change Addendum (v2.1)

01. USER IDENTITY, DETAILS & AUTHENTICATION

முதலில் User-ன் முழு Record, Registration Flow, பாதுகாப்பு மற்றும் Admin-க்கு அனுப்பப்படும் User Identity ஆகியவை வரையறுக்கப்படுகின்றன. Traveler, Hunter, Game அனைத்துக்கும் ஒரே Wayhunter User Account பயன்படுத்தப்படும்.

1.1 User Details / Data Dictionary

Field	Usage / Rule
UserId	Backend உருவாக்கும் immutable, globally unique internal ID; approvals-ல் இணைக்கப்படும். Format owner config.
Display Name / Full Name	Registration name; profile-ல் edit அனுமதி. Admin request-ல் submission-time name snapshot + UserId காட்ட வேண்டும்.
Username	Unique public-friendly profile handle; username change audit where appropriate.
Email / Verified Phone	OTP recovery / contact; uniqueness + verification; contact search தனி consent அடிப்படையில்.
Password Hash	Plain-text சேமிக்கக் கூடாது; confirmation client validation; reset token expiry.
Preferred Language	Tamil, Sinhala, English; Login top-right dropdown; profile setting retained.
Avatar, Bio, Public Location	Profile values, media privacy; private details public disclosure இல்லை.
Account Status	PendingVerification / Active / Suspended / DeletionRequested / Deleted, controlled state changes.
CreatedAt, LastActiveAt	UTC timestamps; user-facing local time formatting.
Device/Abuse Signals	Minimal hashed device binding identifiers where lawful; flag suspicious usage, no automatic blind ban.
Coins & Ledger	Available Coins, Reserved Coins, Lifetime Earned Coins, transaction references; derived server-side.
Level	Earned Coins thresholds; current level and progress; not linked to current spendable balance.
Social / Travel / Hunter / Game	Followers, Following, invites, trips, hunter submissions, games and posts via linked IDs.
Consent & Settings	Contact discoverability, contacts permission, GPS sharing, notifications, privacy, consent timestamps.

1.2 Registration to Home: User Journey

AUTH-UJ-01 புதிய பயனர்

01 App → Language dropdown (Tamil/Sinhala/English) → Create Account.

02 Name + email/phone + password + confirm; validate uniqueness, strength and mandatory fields.

03 Backend provisional UserId உருவாக்கும்; OTP Send → Verify with expiry/attempt/resend cooldown.

04 Verified + active account-க்கு Session உருவாக்கப்படும்; first eligible registration bonus 100 Coins மட்டும் ஒருமுறை வழங்கப்படும்.

05 Onboarding permissions விளக்கம்; contacts permission மற்றும் live GPS permission தேவையான நேரத்தில் மட்டும்.

06 Home: dynamic name/avatar, search, notifications, sponsored banner and Traveler/Hunter/Game cards.

AUTH-UJ-02 Returning User / Recovery

01 Sign In → backend credential/status check → Home.

02 Forgot password → verified recovery token/OTP → new password confirmation → reset and invalidate token.

03 Offline demo login isolated prototype build-ல் மட்டும்; production API-ஐ bypass செய்ய இயலாது.

04 Logout → authenticated session invalidation → Login screen.

ID-001 UserId server-side உருவாக்கப்பட்ட பிறகு மாற்றப்படக்கூடாது; Display Name மாறினாலும் UserId நிலையானது.

ID-002 ஒரே User-க்கு Traveler/Hunter/Game மூன்றுக்கும் தனித்தனியாக கணக்கு தேவையில்லை.

ID-003 புதிய நிறுவல் (download) என்பதன் அடிப்படையில் மட்டும் மீண்டும் 100 Coins தரக்கூடாது; verified first-account activation once only. "Download Bonus" என்பது UI label; enforcement user identity அடிப்படையில். [RECOMMENDED anti-abuse interpretation]

ID-004 Password, OTP, session token, secret keys ஆகியவை Admin request payload-ல் காட்டப்படக்கூடாது.

ID-005 Admin approval queue item எப்போதும் UserId + display name + submission ID + server timestamp + item type கொண்டிருக்க வேண்டும்.

ID-006 UserSuspended என்றால் new submissions/redeems/credits policy-controlled; old records/audit பாதுகாக்கப்படும்.

ID-007 Email/phone uniqueness; OTP expiry, failed attempts, resend cooldown and rate limiting mandatory.

ID-008 Account deletion: identity check, privacy consequences, unspent Coins / pending vouchers treatment policy required before production.

ID-009 Contact and GPS permissions எப்போது வேண்டுமோ அப்போது மட்டும் கேட்க வேண்டும்; permission deny ஆகியாலும் அனுமதிக்கப்படும் மற்ற வசதிகள் இயங்க வேண்டும்.

02. COMMON HOME & PRIMARY NAVIGATION

Scenic Sri Lankan full-screen Home + glassmorphism; Traveler blue/cyan, Hunter gold, Game purple, Profile cyan, Wallet cyan/gold. Home contains dynamic greeting, avatar, search, notification, active sponsored banner, three module cards. Ad rotation 3 seconds in original Admin spec (configurable).

NAV-001 Home / Profile / Friends / Wallet ஆகிய Pages-ல் Global Leaderboard button disabled/hidden ஆக வேண்டும். Game/Hunter/Traveler module-களுக்குள் மட்டும் தத்தம் Leaderboard entry இருக்கும்.

NAV-002 Old primary bottom navigation Home | Hunter | Leaderboard | Profile. "Leaderboard disabled" காரணமாக 4th slot என்னவாகும் என்பது DECISION REQUIRED; disabled control காட்டுவதற்குப் பதில் module-entry வழி பரிந்துரை.

NAV-003 No separate Wallet bottom-nav icon; Profile → My Wallet மற்றும் reward shortcuts வழியாகத் திறக்கும்.

NAV-004 Camera/immersive game screens-ல் Bottom Nav hide செய்யலாம்; safe back path கட்டாயம்.

NAV-005 Error, empty, permission denied, loading and retry states அனைத்து screen-களிலும் இருக்க வேண்டும்.

NAV-006 Tamil/Sinhala/English நீளமான texts மற்றும் குறைந்த screen அளவுகளில் overflow இல்லாமல் responsive layout இருக்க வேண்டும்.

NAV-007 Ad server failure core navigation-ஐ block செய்யக்கூடாது.

03. FRIENDS & FOLLOWERS - UNIFIED PAGE

USER DECISION CH-01. Profile → Followers/Following-ஐ அழுத்தினாலும் Create Trip → Add Friends-ஐ அழுத்தினாலும் ஒரே Friends Page திறக்கும். OpenedFrom context மட்டும் மாறும்.

3.1 Friends Screen Components

Component	Behavior
Header / Tabs	Followers | Following; counts + search; user card, profile and follow/follow-back / unfollow state.
Invite Friends	Wayhunter app download/deep link; native OS share sheet with Gmail, WhatsApp, SMS/normal messages (available apps only).
My Contacts	Explicit opt-in contacts permission; only discoverable/matched Wayhunter accounts list; optional Follow button.
Friend/Follow Requests	Requested / Accepted / Declined / Cancelled statuses; approval when private profile.
Trip context	Create Trip → Add Friends opens same view with "Invite to Trip" action for eligible users; return to existing trip draft.
Empty/Denied	No contacts permission → explanation + retry; no matches → app invite option, never assume contact has app.

FR-UJ-01 Profile → Friends

01 Profile followers/following count tap → unified Friends Page.

02 Select Followers / Following → search → open user public profile / follow or unfollow.

03 Invite Friends → OS share chooser → WhatsApp, Gmail, SMS or other installed apps → send official app link.

04 My Contacts → explain purpose → request permission → compare eligible contacts with registered/discoverable users → Follow request or Follow action.

FR-UJ-02 Create Trip → Add Friends

01 Create Trip draft remains saved locally/in memory → Add Friends opens same Friends Page with trip-selection context.

02 User selects a friend/followed user → Send Trip Invite; invite status pending.

03 Friend accepts → membership valid. GPS live sharing still requires separate explicit opt-in.

04 Return to Create Trip without losing draft; accepted invitees reflected; do not silently share location.

FR-001 Followers and Following are directed relations; a Follow is not automatically a trip invitation.

FR-002 Self-follow, duplicate follow relations, duplicate simultaneous trip invitations and blocked-user invitations prevented.

FR-003 Private profile requires approve/reject for follow request; Public profile can follow per account settings.

FR-004 Contact discovery permission cannot be mandatory for normal app use.

FR-005 Contacts data collection minimized; do not expose nonusers or upload full raw address book without explicit purpose/consent. [RECOMMENDED]

FR-006 Only accounts that permit contact discoverability should appear in matched-contact results. [RECOMMENDED]

FR-007 App link invitation only sends a link; recipients do not automatically become followers or trip members.

FR-008 Sharing channels derive from device OS share options; Gmail/WhatsApp/SMS availability varies by device; no Gmail API connection is required for a native share intent.

FR-009 Follow request send/accept notifications generated once per authoritative state transition.

FR-010 Trip creator cannot add a member into GPS sharing without accepted invitation + separate consent.

FR-011 Blocked users must not see protected profile/private trip content; users can revoke follow/contacts permission.

04. TRAVELER: COMPLETE UJ & VIEW INVENTORY

INHERITED T-01..T-14 + NEW TRAVELER LEADERBOARD. Blue/cyan map-first interface; only live accepted locations/service data shown; historical route reuse remains based on valid completed trips.

ID / Screen	User Action / View	Validation / Result
T-01 Traveler Home	Trusted map places; Create Trip, My Plans, Suggested Plans, AI Trip entry.	Map/search failure retry; correct navigation.
T-02 Create New Trip	Name, dates, start/stops/end, reorder, budget LKR, vehicle, headcount, Add Friends, suggested plans.	Required fields + date validation; Submit → T-05; draft preserved.

ID / Screen	User Action / View	Validation / Result
T-03 Place Picker	Google Map + place search/current-location selection; start/add/end use same picker.	PlaceId/name/GPS and order saved; manual search if permission denied.
T-04 Suggested Plans	Eligible public/shared plans, filters, memories, route, daywise itinerary.	Use/Customize copies draft; original not overwritten.
T-05 My Plans	My/Shared With Me/Completed; edit/share/invite/start.	Start date policy; one active session per trip.
T-06 Start Trip / Live Map	Start, progress, Find, Hazard, Memories, Friends, Pause, I am Safe, Finish.	Explicit Start event and authenticated trip status; GPS/consent controls.
T-07 Find Nearby	On-demand 1 km food/hospital/fuel/police/ATM/attractions.	Results from real provider; category/map/directions; no continuously running search.
T-08 Report Hazard	Active live camera evidence + category + description + GPS/time.	Submit community report; status unverified until verified; back to trip.
T-09 Add Memories	Image/video, caption, tagged trip/place, public/private.	Cloudinary upload authorization, owner permission, return to live map.
T-10 Friends Nearby	Accepted group members, opt-in live distance, live/paused/offline and I'm Safe.	No unconsented GPS; show stale/paused honestly.
T-11 Finish Trip	Confirm actual session end, visited places, memories, route summary.	Only actual Started → Finished completion counts; stop sharing.
T-12 Share Trip / Add Friends	Open unified Friends Page and issue trip invitations.	Accepted member =/location consent; pending/accepted/declined states.
T-13 AI Trip Planning	Date, locations, budget, vehicle, group, preferences; Gemini or approved external API.	Editable daywise proposal; estimate labels; API error preserves draft.
T-14 Completed History	Actually completed trips with valid start/end; View, Reuse Plan.	Reused plan new draft; no fabricated route.
T-15 Traveler Leaderboard [NEW]	Rank by count of actually Finished Trips; public profile, own position.	Save/Submit alone does not count; suspicious/noneligible completion excluded.

TR-UJ-01 Trip Creation → Completion

01 Home Traveler → T-01 → T-02; pick start, ordered stops and end via T-03.

02 Optional Suggested T-04 / AI T-13 → copied or generated editable trip draft.

03 Add Friends opens Friends page → optional Trip Invite; save/submit trip plan to T-05.

04 Before planned start date/time send reminder to eligible user; the reminder NEVER starts the trip.

05 User explicitly presses Start → backend records StartedAt, status Active and active session.

06 Active trip enables live map, on-demand nearby, hazard reports, memories and consent-based group GPS.

07 User presses Finish Trip → backend validates Active trip + end event, saves FinishedAt and Completed status.

08 Only valid Completed trip counts for Traveler Leaderboard and historical route segments; group live sharing stops.

TR-UJ-02 Hazard Report → Other User Pop-up

01 User A in active trip submits hazard with GPS, photo/video and server timestamp; initial Community Report state.

02 Server stores hazard, suitable expiry/coverage/direction metadata, and moderation queue.

03 User B on an active trip approaches the affected route/area: server/client evaluate valid hazard proximity and direction/recency.

04 User B receives context-aware pop-up with category, approximate location, freshness and verification label; dismiss/view safer route.

05 Resolved/expired/invalid reports stop being shown as active warnings; admin audit retained.

TR-001 "Create/Submit Trip" never marks it Completed and never yields Traveler ranking points.

TR-002 Trip completion requires authenticated Start Trip and explicit valid Finish Trip events; server saves both timestamps.

TR-003 Planned start date reminder only; user must manually begin trip; reminder timing is owner-configurable.

TR-004 No duplicate active session for same trip; restart/reconnect policy restores previous session instead of generating false completion.

TR-005 Actual route/visit evidence is not fabricated when GPS unavailable; show "route unavailable" if missing.

TR-006 Only completed authorized shared trips and ordered consecutive same-trip segments may feed historical route recommendations.

TR-007 User can pause/stop live sharing and I am Safe; trip finish revokes live sharing.

TR-008 Friends Nearby only accepted, explicitly opted-in participants; private trip content cannot be shared through public plan card.

TR-009 1 km Find only when clicked and with valid location; category filters and provider error state.

TR-010 Hazard creation does not require routine manual photo approval; report marked unverified pending authorized confirmation.

TR-011 Hazard alert pop-up should show unverified label when unverified, and must not be presented as guaranteed emergency information.

TR-012 Hazard proximity, direction, age and repeat-warning cooldown are owner-configured; do not spam users each GPS update. [RECOMMENDED]

TR-013 Only active, unexpired hazards relevant to the moving user should trigger automatic pop-ups.

TR-014 AI plan is suggestion; all budget/time/place estimates editable and clearly approximate.

TR-015 Reuse original trip must never mutate source trip; use editable copied plan.

05. LOCATION HUNTER: COMPLETE UJ & RULES

USER DECISION CH-02 overrides Admin PDF auto-verification. Every Hunter Place Submit goes to Admin Queue with UserId + Name and remains Pending until explicit Approve or Reject. Mandatory live camera capture only; gallery is prohibited for proof.

ID / Screen	User Action / View	Validation / Result
H-01 Hunter Home	GPS district, search, existing discoveries, My Hunts, Add Place and Hunter Leaderboard.	Only current GPS district enabled for submitting.
H-02 Add New Place / GPS	Name, description, current district, GPS accuracy; start evidence.	GPS mandatory, no manual coordinate override.
H-03 Guided 360 Recording	Live capture with guide, timer/rotation, cancel/retry.	Gallery disallowed; guided sweep must not be called spherical 360.
H-04 360 Preview	Playback, Retake, blue Add Normal Photo.	Invalid 360 capture must be retaken.
H-05 Live Normal Photo	Camera proof, GPS/time status.	No gallery photo selected.
H-06 Photo Preview	Retake / Confirm.	Reject corrupt/empty files.
H-07 200m Duplicate Check	Nearby candidates, name/category/image similarity and distance.	200m screening only; not auto-reject by distance.
H-08 Review & Submit	Place data, GPS, 360, photo, flags, Submit.	Send UserId + Name + SubmissionId to Admin; Pending until decision.
H-09 Verification Status	Pending / Approved / Rejected and reason/timestamps.	Owner sees status and admin response; pending no Coins.
H-10 Reward Summary	Congratulations + 200 Coins upon eligible admin approval.	Issue exactly once; no reward on rejection.
H-11 Existing Discoveries	Map/list of published approved places with search/district filters.	No pending/rejected public locations.
H-12 Place Details	Approved place, public photos, description, directions, verified badge.	Private evidence not shown.
H-13 My Hunts	Own places with Pending / Approved / Rejected; status + detail/reason.	Show all statuses to owner; avoid repeated submission reward.
H-14 Hunter Leaderboard	Rank by total approved Hunter-earned Coins.	Only confirmed contributions, duplicate reward excluded.

H-UJ-01 Submit → Admin → My Hunts → Coins

01 Hunter Home → Add Place; GPS district/accuracy validated.

02 Live 360 video → preview → live normal photo → preview; no gallery option.

03 200m candidate screening, restricted-zone and evidence checks → Review and Submit.

04 Backend creates unique SubmissionId and AdminQueue record (UserId, Name, coordinates, place, evidence, flags, timestamp).

05 My Hunts displays Pending; Admin reviews all new submissions, then Approve or Reject with reason.

06 Approve → publish place + credit 200 Coins exactly once + congratulations notification + My Hunts Approved.

07 Reject → no Coins; My Hunts Rejected + reason/notification; audit retained.

HU-001 Hunter proof must be live in-app 360-guided video and live normal photo: gallery upload for either is prohibited.

HU-002 User cannot edit reported GPS coordinates; accuracy/time and session linkage must be recorded.

HU-003 Only current detected district open for submission; off-GPS shows permission/help/retry.

HU-004 200m duplicate radius is candidate discovery, not automatic rejection.

HU-005 Restricted/military/railway/blocked zones trigger rejection/escalation under configured safety rules.

HU-006 Every new Hunter submission pending Admin; auto-publish disabled under revised user rule.

HU-007 Every Admin review item contains UserId and UserName as required; privileged reviewers also see evidence and flags.

HU-008 Admin Approve/Reject decisions include actor/time/mandatory reason for rejection and audit.

HU-009 200 Coins credited after successful eligible approval only, once per submission (idempotent).

HU-010 Re-submitted/rejected duplicate place cannot farm new coins without distinct newly approved eligible submission.

HU-011 My Hunts shows Pending/Approved/Rejected only to owner; public map shows Approved & Published.

HU-012 Approval success message combines “Congratulations” + credited Coins; rejection shows reason without false reward.

HU-013 Restricted evidence storage remains private even after public place approval.

06. GAME: COMPLETE UJ & RULES

USER DECISION CH-03. Game create Submit → Admin Review (Pending) → Approve/Reject. Game completion proof also Admin Approved before 100 Coins. Public game available one month; after expiry hidden from public but retained in creator’s My Games as Expired/Archived. The Admin source had conflicting auto-publish; it is superseded here.

ID / Screen	User Action / View	Validation / Result
G-01 Game Home	Create Game, Play Game, My Games, Rules, Game Leaderboard, Rewards.	Module-only leaderboard accessible.
G-02 How to Play	Hints, live proof, timer, no dangerous/trespassing tasks.	Must clearly explain review/rewards policy.
G-03 Play Game List	Approved, Published and unexpired games only.	No pending/rejected/expired availability.
G-04 Game Details	Creator, game title, stages, clock, reward, Start.	Check not own game; active attempt eligibility.
G-05 Hint Stage	Photo + text clue, stage timer and progress.	No compulsory map/km hint; no answer leak.
G-06 Player Live Camera	Take live proof against stage.	No gallery proof; associate time/session/context.
G-07 Proof Review	Retake/Submit; Pending/Approved/Rejected.	Accepted server/admin checks before next stage/final credit.
G-08 Victory	Completion confirmation and 100 Coin pending/approved result.	Do not award Coins before admin approval.
G-09 Game Over	Timeout/failure; eligible retry.	No 100 Coins for failed/unverified completion.
G-10 Create Game	Title/description/cover, duration/reward, stages.	Save draft then admin submission.
G-11 Photos & Hints	Creator uploads clue photos and text; reorder stage list.	Creator gallery allowed (unlike player live proof).
G-12 Preview & Submit	Final preview, Submit for Admin Approval.	Pending immediately; no public play before approve.
G-13 My Games	Created: Draft/Pending/Approved/Rejected/Active/ Expired; Played: In Progress/Completed/Pending Result.	Creator sees expired archived games; player sees completed history.

ID / Screen	User Action / View	Validation / Result
G-14 Game Leaderboard	Ranking by cumulative approved Game-earned Coins.	No fake/pending/reversed credits.
G-15 Rewards Store	Offers and Coins requirement; redirect shared Wallet redemption.	Show partner validity and stock.
G-16 Game History	Played date, stages, status, final proof/Coins.	Owner only where private; archive preserves history.

G-UJ-01 Create Game

01 Game Home → Create Game → title/cover/description/time and stages with photo/text hints.

02 Preview → Submit → backend creates GameId + UserId + CreatorName and status PendingAdmin.

03 Admin reviews safety, clues, policy, anti-cheat and reward conditions → Approve or Reject with reason.

04 Approve → PublishedAt, ExpiresAt based on one-month policy, status Active; credit Creator 100 Coins exactly once and notify.

05 Reject → not public, no creator Coins, show reason in My Games.

06 When expired → remove from public Play Game List automatically; creator My Games still shows record as Expired/Archived.

G-UJ-02 Play and Complete

01 Player opens Approved active game, sees photo/text hints + timer; owner cannot play own game.

02 Player captures proof in-app live camera; proof validated; each stage moves only when accepted.

03 Final completion creates PendingAdminVerification result with Player UserId+Name, GameId, session/proof and time.

04 Admin verifies eligible completion / winner against anti-cheat policy; Approve awards 100 Coins exactly once; Reject awards none.

05 My Games → Played/Completed displays final status; Game Leaderboard uses only approved Game Coins.

GA-001 Game creation must be explicitly approved by Admin; auto-publication disabled for new game under user revision.

GA-002 Creator receives 100 Coins on approved published game once; drafts/pending/rejected games earn 0.

GA-003 Player receives 100 Coins only for completed game with admin-approved verified result; pending/failed/rejected earns 0.

GA-004 Normal community game source rule: first valid verified winner wins; creator cannot play their own game. Multi-winner policy is DECISION REQUIRED for 100 Coins per finisher.

GA-005 Creator hint images may be selected/uploaded from library; player proof must be live camera only.

GA-006 No mandatory map or km hints; keep staged photo and text clues, safety instructions and timer.

GA-007 Public availability lasts one month as specified. Clock anchor (approval/publish date) and calendar-month semantics require confirmation; draft proposes publish/approve time. [RECOMMENDED]

GA-008 Expiry automatically disables public discovery/start; keep immutable creator ownership and My Games archived record; do not physically delete creator history.

GA-009 If game expires while already in progress, attempt treatment requires explicit policy; no unverified late reward. [DECISION REQUIRED]

GA-010 Approval queue contains GameId, Creator UserId, UserName, details, stages, evidence, timestamps; game result queue adds Player UserId+Name.

GA-011 Rejection reason shown to owner; credits and status change idempotently; audit every decision.

GA-012 Pending and expired games not visible for public play; suspended/flagged games immediately blocked.

GA-013 Anti-cheat includes session/time, duplicate media, GPS anomalies when applicable; do not require undisclosed creator map.

07. THREE MODULE LEADERBOARDS

Leaderboard	Sort Metric	Navigation / Key Rule
Game	Approved cumulative Coins from Game creation/play only, descending.	Game module only; reject pending/duplicate/reversed.
Hunter	Approved cumulative Coins from approved Hunter places only, descending.	Hunter module only; 200 Coins per eligible place.

Leaderboard	Sort Metric	Navigation / Key Rule
Traveler	Number of trips actually Started and Finished, descending.	Traveler module only; Submit/save alone never counts.

LB-001 Global Leaderboard button disabled/hidden on Home, Profile, Friends and Wallet.

LB-002 Ranking query server-side; expose public-eligible display name/avatar and own rank, not private wallet or GPS.

LB-003 Define ties consistently using earliest qualifying verified achievement then UserId as stable tiebreaker. [RECOMMENDED]

LB-004 Exclude reverted fraudulent reward credits, banned users where approved policy, and ineligible trips.

LB-005 Period filter (all-time/monthly) not defined by user; default cumulative all-time pending owner approval.

LB-006 When points equal use explicit tie display or stable ordering; do not silently claim distinct merit.

08. PROFILE & SETTINGS: P-01 .. P-18

ID / Screen	User Action / View	Validation / Result
P-01 My Profile	Scenic header, avatar, bio, followers/following count, badges/level star, Games/Posts/Trips/Hunt/My Wallet tabs.	Numbers and account states live from backend.
P-02 Edit Profile	Name, unique username, photo, bio, location.	Owner only + validation.
P-03 Games Tab	Created/played/won game summaries + See All.	Use G-13/G-16; no pending counted as verified wins.
P-04 Post Details	Photo/video/caption/place, share/delete.	Privacy and ownership.
P-05 Trips Tab	Upcoming/completed trips.	Route details and private trip rules.
P-06 Hunt Places Tab	Own submitted Hunts and status.	Link H-13 My Hunts, including pending/rejected.
P-07 Followers/Following	Unified Friends Page entry and counts.	Same screen as Traveler Add Friends.
P-08 Badges & Level	Star Level, 2000 cumulative-earned Coins per level and progress bar.	Level unaffected by Coins redeemed; badge requirements configured.
P-09 Other User Profile	Public bio, posts, eligible trips/places and Follow.	No private data/wallet shown.
P-10 Settings Home	Security, Language, Privacy, Notifications, Help, About, Delete Account, Logout.	Preserve selection.
P-11 Security & Password	Old/New/Confirm password.	Server verifies, secure update.
P-12 Language	Tamil/Sinhala/English dropdown.	Persist + fallback.
P-13 Privacy	Public/Private account.	Private follow approval and visibility.
P-14 Notifications	Toggle trip, friends, Hunter, Game, Coins, social, safety, promotions.	Honor OS permission, save preference; critical in-app safety availability.
P-15 Help & Support	Guides, FAQs, contact, report issue.	Tracked support request.
P-16 About	Version, update info, terms and policy.	Only verified version/update claims.
P-17 Delete Account	Yes/No + authorized identity check.	Retention / vouchers / pending redemption policy.
P-18 Logout	Confirm/Cancel bottom sheet.	Confirm revokes session and returns login.

Original 18 P-views தொடர்கின்றன. XP/HP/RC தகவல்கள் அனைத்தும் Level/Coins-க்கு மாற்றப்பட்டுள்ளன. Profile-ல் Wallet Option மட்டும்; global leaderboard shortcut வேண்டாம்.

09. MY WALLET: COINS ONLY

USER DECISION CH-05. பழைய HP, RC, Lifetime HP, XP அனைத்தும் நீக்கப்பட்டு ஒரே "Coins" மட்டுமே. Level star மற்றும் straight progress bar W-01 Wallet Home-ன் மேல்பகுதியில் இருக்கும்.

9.1 Coins rules

Action	Coins	Approval/Rule
First verified account / app welcome	+100	Only once per eligible account; download bonus interpreted as server-verified first activation.
One approved Hunter discovery	+200	Admin approves; congratulations notification; one credit per valid place.
One approved created game	+100	Admin approves/publishes; one credit per created game.
One approved completed game	+100	Admin validates eligible completed play/win; once per eligible result.
Coin cash reference	10,000 Coins = LKR 100	Conversion factor LKR 0.01 per Coin; actual payout channel/eligibility terms pending.
Level threshold	2,000 lifetime earned Coins / level	Current Level = floor(lifetimeEarned/2000)+1 [proposed], progress = remainder.
Minimum redemption request	10,000 Coins	"10,000-க்கு மேல்" vs "10,000 அல்லது அதற்கு மேல்" requires final decision; provisional >=10,000.

9.2 Wallet Screen Register

ID / Screen	User Action / View	Validation / Result
W-01 Wallet Dashboard	Available Coins, Total Earned Coins, Reserved Coins, star Level and straight progress bar /2000; Redeem Rewards, My Vouchers, Wallet Rules.	Ledger-authoritative; progress based on lifetime earned.
W-02 Redeem Rewards	Coins balance + LKR equivalent box + eligible reward/payout choices.	Minimum and offer terms; ask admin approval.
W-03 Confirm Request	Coins amount, equivalent LKR, expected remaining balance, confirm/cancel.	Server checks balance and duplicate request; pending request creates hold.
W-04 Request Status	Pending/Under Review/Approved/Rejected/Cancelled/Fulfilled.	Pending creates a Coins reservation, not a posted debit. Final debit on Admin Approval versus on verified Fulfillment is VCH-DEC-01 DECISION REQUIRED. Rejected/cancelled requests release reservation.
W-05 My Vouchers	Pending requests and approved/issued/used/expired vouchers with amount and expiry.	Only issued voucher usable; do not show fictional voucher code.
W-06 Voucher Details	Code/QR if issued, merchant, amount, expiry and usage terms.	Unique validation; owner-only.
W-07 Wallet Rules	Coins earn amounts, exchange reference, request steps, eligibility and disclosures.	Display latest admin-configured values + change notice.
W-08 Coins Transaction History	Welcome, Hunter, Game, redeem hold/approve/release, voucher issue/use, corrections.	Append-only audit; timestamp/source ID/status, only verified credits.

W-UJ-01 Redeem Request

01 User opens Profile → Wallet → sees Coins, total earned, Level star and progress.

02 Click Redeem Rewards; box shows redeemable Coins and corresponding LKR reference or available voucher offer.

03 At least configured threshold and sufficient Available Coins → Confirm creates Pending Admin request with UserId+Name.

04 Visible Total Balance is NOT permanently debited during Pending; internally reserve requested Coins to prevent second spending.

05 Admin approval and voucher fulfillment are distinct events. FINAL Coins debit trigger (on approval vs confirmed fulfillment) is DECISION REQUIRED, ref VCH-DEC-01; until resolved, no production debit should be enabled. Pending Coins stay reserved; user is notified of confirmed state.

06 Admin rejects/cancels → release reservation; coins remain available; reason/notification and audit.

07 Voucher issued → My Vouchers shows amount, expiry and code/QR only if actually assigned; used/expired changes recorded.

W-UJ-02 Level Progress

01 App awards one-time +100 welcome on eligible registration and records cumulative total.

02 Every approved earned event increases lifetime-earned Coins and Level progress; pending events do not.

03 Every 2,000 lifetime-earned Coins completes progress → increase next star level; bar resets to remainder.

04 Redeeming Coins lowers spendable balance but not lifetime-earned level points; fraud corrections must adjust valid lifetime-earned total as policy.

CO-001 Only one Coin currency is allowed in new UI and accounting; remove all old HP/RC/XP and their exchange logic.

CO-002 Available Coins, Reserved Coins and Lifetime Earned Coins are distinct accounting fields under one currency, not additional currencies.

CO-003 Wallet shows current Available Coins and Total Coins Earned separately so user does not mistake historical total as spendable.

CO-004 100 welcome Coins once per eligible verified account, never per reinstall/reset/new device.

CO-005 Approved Hunter +200 Coins; Approved Game Creation +100; Admin-approved eligible Game Completion +100; all idempotent.

CO-006 Pending or Rejected Hunter/Game cannot award Coins.

CO-007 Level is cumulative earned coins in increments of 2,000. Spending must not reduce level. [RECOMMENDED accounting interpretation]

CO-008 10,000 Coins = LKR 100 is a conversion reference; direct cash payout cannot be promised without approved payout channel and settlement rules.

CO-009 User stated “above 10,000” and “10,000 Coins = Rs.100”; exact redemption boundary >= or > 10,000 is a final DECISION REQUIRED.

CO-010 Pending redemption must not permanently decrease displayed Total Coins; however spendable portion is reserved server-side to block double requests.

CO-011 If Pending request is rejected or cancelled, release reserved amount without duplicate credit.

CO-012 [REVISED / DECISION REQUIRED] Admin Approval vs confirmed Fulfillment as the final Coins debit trigger is disputed across BRD v2.0 and Admin Working Process pp.14,19. Pending request must retain reservation; final debit must happen once with inventory/fulfillment safeguards after Product Owner approval of VCH-DEC-01.

CO-013 Voucher fulfillment requires genuine unique code/QR, owner assignment, expiry date and amount; Pending/Approved-but-unissued is not usable.

CO-014 Voucher payout amount or discounts follow actual supplier agreement; fixed exchange reference does not guarantee a merchant voucher face value.

CO-015 Show applicable rules, transaction history and request status to user; all wallet mutations via backend ledger.

CO-016 No casual manual adjustment in Admin User List; corrections require reason, approval and compensating ledger entry.

CO-017 Never credit/redeem the same request/reward twice on retries or network timeouts.

10. NOTIFICATIONS, HAZARD POP-UPS & SCHEDULING

Event	Recipient	Required behavior
Follow Request Sent	Target user	In-app/push alert, Accept or Decline, request reference.
Follow Request Accepted	Requester	Acceptance notification; user relationship updated.
Trip Invite / Accepted	Inviter and invitee	Invitation/reminder state; live GPS still opt-in.
Trip Start Reminder	Trip creator/member as eligible	Configured time before planned trip start; never auto-start.
Hunter Pending/Approved/Rejected	Submitting Hunter	Status update; Approved + congratulations +200 Coins; rejected reason.

Event	Recipient	Required behavior
Game Creation Pending/Approved/Rejected	Game creator	Published/created reward +100 only after approval.
Game Proof Pending/Approved/Rejected	Game player	Completion reward +100 only after admin approval.
Redeem Pending/Approved/Rejected	Redeeming user	Pending status; Coins settlement on approval; reason on rejection.
Active Route Hazard	Nearby active traveler	In-trip hazard pop-up on relevant route while report active; label verification state.
Voucher Issued/Expiry	Voucher owner	Code available on issue; expiry reminder where configured and opted in.

NOT-UJ-01 Trip Start and Hazard

01 Planner submits future trip → planned date/time stored; notification scheduling status saved.

02 Before planned start (exact offset configurable) send "Trip starting soon" notification once or per approved reminder policy.

03 User must explicitly press Start Trip. Only then track active route for contextual hazard detection.

04 On matching active hazard from another user, show clear pop-up with details and Reported/Verified state.

05 User may dismiss/view relevant route; on finish, automatic in-trip location monitoring and sharing stop.

NO-001 Send requests/approvals based on real persisted state transitions, not an optimistic UI button click alone.

NO-002 Push Notification and In-App History should retain event reference/status without exposing sensitive exact private GPS in lock-screen text.

NO-003 User can manage category preferences; critical safety info remains visible in-app regardless of promotional push choice.

NO-004 Do not send outdated hazard pop-up for Resolved/Expired/Flagged invalid reports.

NO-005 Hazard reports are community submissions, not verified emergency guarantees unless verified.

NO-006 Only active trip location/session should power route hazard detection; collect minimal location and stop on Finish.

NO-007 Notification delivery can fail; transaction/reward must still persist and be visible after reopen.

NO-008 Deduplicate notifications across retries; record sender/system, recipient, target, time and delivery status.

NO-009 No unapproved hotel-offer promotional push; source scope excluded it.

NO-010 Reminder time, hazard radius/age/cooldown/priority and vouchers expiry reminders require configurable policy.

11. ANGULAR ADMIN PORTAL - 17 MODULES & ADMIN USER JOURNEYS

Admin baseline 17 modules from supplied 20-page specification. All HP/RC columns, minimum thresholds, summary cards and reports must be converted to Coins. All new Hunter/game approvals are manual; source auto-approval language superseded.

ADM-01 Admin Login

Main screens & data: Only authorized Admin login; email/password/show; Forgot/reset with OTP; role validation, no public sign-up.

Admin UJ: Open admin → login → server validates role/status → Dashboard; failure shows safe generic error.

Key rules: Admin credentials/status secure; login attempts/session timeout/audit.

ADM-02 Dashboard

Main screens & data: Total Users, Published Places, Active Games, Pending Hunter & Game Approvals, Active Hazards, Merchants, Coins Liability, pending redemption, activity map, analytics, ads.

Admin UJ: Login → dashboard → select pending Hunter/Game/Coins request by UserId + Name → appropriate review module.

Key rules: Dashboard monitoring not arbitrary balance edit; charts must use Coins only.

ADM-03 User Management

Main screens & data: Search UserId/name/email/phone/status/district; profile, friend request counts, trips/hunts/games/Coins summary, reports, devices; Warn/Suspend/Reactivate.

Admin UJ: Users → filter → User Details → choose action → enter reason → confirm → audit + appropriate notification.

Key rules: No private GPS browsing without need; do not directly edit Coins; UserId immutable.

ADM-04 Hunter Verification

Main screens & data: Every new submission Pending; list UserId+Name/SubmissionId, district/GPS accuracy, live 360/photo, 200m similarity, restricted flags; Approve/Reject/Request Evidence/Flag.

Admin UJ: Queue → open submission → examine proof/duplicate/safety → approve + publish +200 Coins OR reject with reason → notify → My Hunts status.

Key rules: Manual approval for all new Hunts overrides older auto verified flow; audit, one award.

ADM-05 Advertisement Management

Main screens & data: Create/edit/archive Home banners + Traveler/Game/Hunter placements; sponsor, target district, schedule, display order, 3-second rotation, views/clicks.

Admin UJ: Ads → create banner → target and dates → preview → publish → mobile active ads → monitor.

Key rules: Expired/paused not active; sponsor label; no verification bypass.

ADM-06 Places Management

Main screens & data: Approved published place list/detail, official history, then/now media/sources; edit/deactivate/merge duplicate.

Admin UJ: Places → choose approved place → edit/merge/archive with reason → propagate to mobile.

Key rules: Raw pending Hunter submission remains ADM-04; preserve memories/history on merge.

ADM-07 Trips & Historical Routes

Main screens & data: Trip statuses with actual StartedAt/FinishedAt, ordered stops, shared plans, historical consecutive segments and invalid route flags.

Admin UJ: Trips → inspect actual completed session → validate ordered stop segments → flag ineligible data → save audited decision.

Key rules: Submit-only trips never complete/rank; reuse only eligible completed consecutive segments.

ADM-08 Hazards

Main screens & data: Reported/Unverified/Verified/Resolved/Expired states; GPS and time, severity, evidence, map/timeline, verify/resolve/expire/flag.

Admin UJ: Hazards → inspect evidence/location/age → review → change state with note → push updated hazard status.

Key rules: Reporter cannot self-verify; expired does not alert; preserve audit.

ADM-09 Group Trips

Main screens & data: Active/completed/flagged trips, organizer, invitation/member status, explicit GPS consent, separation alerts, I am Safe, support actions.

Admin UJ: Group trips → select flagged/active → safety alert → authorized intervention → audit.

Key rules: Admin sees minimum safety-relevant location; stop sharing when trip ends.

ADM-10 Game Management & Anti-cheat

Main screens & data: Creation review queue UserId+Name/GameId/stages/timer/safety; Player completion approval queue UserId+Name/session/proof; active/expired/flagged, first winner, month expiry.

Admin UJ: New Game Submit → manual approve publish +100 creator Coins; final Play Proof → approve verified eligible result +100 player Coins; reject no credit; expiry archives public.

Key rules: No auto publish; creator cannot win own game; idempotent award; reason for rejection.

ADM-11 Coins & Rewards

Main screens & data: Search wallet by UserId/name; Available/Reserved/Lifetime Earned, level progress, Coin ledger, reward sources, fraud review, correction entries and rule configuration.

Admin UJ: Coins → search user → inspect sources → review suspicious credit → controlled reversal with reasons/audit.

Key rules: Old HP/RC/XP removed; no casual balance edit; one-currency ledger.

ADM-12 Merchant / Partner Management

Main screens & data [EXTENDED]: Partner directory, agreements, offer stock/code ranges, branches/contact, validity/expiry, delivery, voucher source image/PDF upload and protected preview, manually verified code/amount/expiry, optional OCR and QR/barcode assisted extraction, optional official supplier API.

Admin UJ [EXTENDED]: Create partner/offer → upload optional source image/PDF OR manual code entry → validate fields/source/uniqueness/expiry and supplier agreement → authorized Admin verifies record → save genuine voucher stock → publish eligible offer → monitor. OCR/scan assisted entry is PROPOSED only.

Key rules: No mandatory merchant portal; official partner API only if available; no guaranteed voucher acceptance without written agreement.

VCH-ING-01 [IN SCOPE CLARIFICATION]: Voucher image/PDF upload and manual entry require MIME/size validation, private media storage, preview, protected voucher codes, source traceability, explicit verifier identity/time and uniqueness checks before stock becomes Available. Source images alone are not proof of genuine merchant validity.

VCH-ING-02 [PROPOSED / NOT APPROVED]: OCR extraction and QR/barcode prefill may assist the Admin but shall never auto-approve inventory. Scanner support, implementation cost and acceptance criteria require Product Owner decision VCH-DEC-02.

ADM-13 Redemptions / Settlements

Main screens & data: Pending/Approved/Rejected/Fulfilled requests with UserId+Name, Coins held/debited, payout vs voucher type, stock, code allocation, partner settlement/status and audit.

Admin UJ [REVISED]: Open pending → verify active Coins hold and valid voucher stock → approve/reject with reason → allocate a unique voucher → issue/fulfill → finalize a single debit at the PRODUCT-OWNER-APPROVED trigger (approval or fulfillment; VCH-DEC-01 unresolved). Reject/cancel → release hold. Audit every step.

Key rules: Never permanently deduct during pending; partner or cash payout only under approved delivery method; safe reversal.

ADM-14 Campaign Management

Main screens & data: Sponsored games, district passes, verified missions, promoted places, dates, audience, sponsor, Coins budget, clear disclosure.

Admin UJ: Campaign → choose type/sponsor → dates/eligibility/Coins budget → preview → schedule/publish → monitor → archive.

Key rules: Sponsored reward cannot bypass Hunter/Game approval or duplicate Coin payout.

ADM-15 Reports & Analytics

Main screens & data: Users, travel completed count, Hunter approved count, game created/won, pending approvals, Coins minted/redeemed/reserved, voucher stock, hazards, ads revenue.

Admin UJ: Analytics → domain/date/district/status filters → charts/record drilldown → authorized CSV/PDF export.

Key rules: Avoid private GPS exposure; use verified ledger; old HP/RC KPIs retired.

ADM-16 Notifications

Main screens & data: System/admin composer and scheduled sends, target roles/districts, FCM delivery/history; new social, trip, approval, Coins and hazard event templates.

Admin UJ: Compose → target → preview → send/schedule → record audit; automatic backend events use template and user preferences.

Key rules: No private data in push text, no unapproved hotel promotional pushes; bulk sends require confirmation.

ADM-17 Settings, Restricted Zones, Audit & Admin Mgmt

Main screens & data: Languages, media limits, 360/camera rules, 200m radius, 30-day game expiry policy, 2000 Coin level, Coin earnings, ≥10k redeem threshold, admin roles, API status, device binding.

Admin UJ: Authorized admin → configuration → validate and confirm sensitive change → apply versioned rule → audit.

Key rules: Only permitted roles alter economic/safety rules; secrets not logged/plain displayed; no public admin registration.

11.1 Approval Queue - Shared Layout

Field / Control	Hunter Queue	Game Queue	Redemption Queue
Identity	UserId + UserName	Creator/Player UserId + Name	Requester UserId + Name
Reference	SubmissionId, PlaceId	GameId, AttemptId	RequestId, OfferId
Evidence	GPS, district, live photo, 360, duplicate flags	Clue content or player live proof + session	Coins amount, reservation, stock, terms
Status	Pending/Approved/Rejected	Pending/Approved/Rejected/ Expired	Pending/Approved/Rejected/ Fulfilled
Actions	Approve, Reject, Request Evidence, Flag	Approve, Reject, Flag/Disable	Approve, Reject, Allocate, Fulfill
Effects	Publish +200 Coins or no credit	Creator +100 or player +100 Coins; expiry archive	Final Coins debit trigger NOT DECIDED (Admin Approve vs confirmed Fulfill). Reserve during Pending; release on Reject/Cancel; allocate/issue verified voucher only.
Audit	Admin ID, timestamp, reason, version	Admin ID, timestamp, reason, proof	Admin ID, timestamp, ledger references

12. GLOBAL BUSINESS RULE CATALOG

அனைத்து Cross-module நுணுக்கமான விதிகளும் ஒரே இடத்தில் மீண்டும் சுருக்கமாக குறிப்பிடப்படுகின்றன; முன் பகுதிகளில் உள்ள விளக்கங்களை ஒவ்வொரு development test-க்கும் இணைக்க வேண்டும்.

SYS-001 ஒவ்வொரு entity-க்கும் server-generated unique immutable ID; CreatedAt/UpdatedAt and audit metadata.

SYS-002 All business-state transitions server-side; Flutter/Angular UI cannot self-approve or self-credit.

SYS-003 Repeat API requests must be idempotent for registration bonus, Hunter submit/approval, Game creation/play approval, trip finish, Coins redeem and voucher allocation.

SYS-004 Admin actions: authenticated, authorized, reason where sensitive, timestamp, affected UserId/item, before/after state and auditable.

SYS-005 Notifications and live updates eventually consistent; status source-of-truth remains backend database.

SYS-006 Existing retired HP/RC/XP old data must have audited migration plan before launch; do not silently convert with invented rates.

SYS-007 Non-sensitive public profile searchable; private accounts have restricted follower/trip/media visibility.

SYS-008 Consent revocation must actually stop further live sharing/contact discovery; stale cached views updated.

SYS-009 Age/eligibility/legal terms for cash or voucher redemptions must be reviewed before public release.

SYS-010 Monetary calculations require exact decimal representation and clear denomination; avoid client floating point authority.

SYS-011 All rule values that could change must be centrally configured/versioned, with historical transaction terms preserved.

SYS-012 Error/retry path must not duplicate rewards, starts/finishes, follower actions or vouchers.

13. STATE MACHINES & DATA REQUIREMENTS

Entity	Statuses & permitted transitions
User	PendingVerification → Active ↔ Suspended; Active → DeletionRequested → Deleted under policy.
Follow Request	Pending → Accepted/Declined/Cancelled; cannot accept twice.
Trip	Draft → Submitted/Planned → Active (Start event) → Completed (Finish event); Cancel/Flag exceptions; no direct Planned → Completed.
Trip Invite	Pending → Accepted/Declined/Cancelled; membership after accepted; GPS consent separate.
Hunter Submission	Draft → PendingAdmin → Approved & Published OR Rejected; RequestEvidence/Flag exception, logged.
Game Creation	Draft → PendingAdmin → Approved/Active OR Rejected; Active → Expired/Archived after month; Flagged/Disabled exception.
Game Attempt	Started → StageProofs → PendingAdminFinal → ApprovedCompleted OR Rejected/Failed/Expired.
Coins Redeem	Created → Pending & Reserved → Approved → Assigned → Issued/Fulfilled; or Rejected/Cancelled & Released. Placement of Debited transition is VCH-DEC-01 Decision Required.
Voucher	Unassigned → Assigned/Issued → Used OR Expired/Revoked; owner-only code access.
Hazard	Reported/Unverified → Verified → Resolved/Expired; invalid flagged; status changes audit.
Notification	Queued/Scheduled Sent/Delivered/Failed/Read with deduped event reference.

13.1 Proposed SQL Domain Entities (conceptual - not final ERD)

Area	Important tables/entities
Identity	Users, UserProfiles, Sessions, OtpChallenges, UserDevices, UserPermissions, UserConsents, PrivacySettings
Friends	Follows, FollowRequests, ContactDiscoverabilityConsents, ContactMatchEvents, UserBlocks
Travel	Trips, TripStops, TripSessions, TripMembers, TripInvitations, TripShareConsents, TripLiveStatus, TripHistorySegments

Area	Important tables/entities
Safety & Media	Hazards, HazardConfirmations, HazardAlerts, MediaAssets, MediaPermissions, Memories
Hunter	PlaceCatalogue, HunterSubmissions, SubmissionEvidence, DuplicateCandidates, HunterDecisions
Games	Games, GameStages, GameClues, GameReviews, GameAttempts, GameProofs, GameCompletionDecisions, GameExpiries
Coins	CoinWallets, CoinTransactions, CoinReservations, CoinRewardSources, CoinLevels, CoinRuleVersions
Voucher	Partners, PartnerOffers, VoucherInventory, RedemptionRequests, VoucherAssignments, VoucherUses, Settlements
Operations	Notifications, NotificationDeliveries, Ads, Campaigns, AdminUsers, Roles, AdminPermissions, AuditEvents, Configurations, RestrictedZones

13.2 Referential & Transaction Safety

- UserId is shared foreign key for Hunter, Game, Wallet and approvals; submission and review also store a display-name snapshot so admin review remains understandable after username changes.
- Reward source has unique constraint on (RewardType, SourceId, RecipientUserId), preventing duplicate approval credits.
- Coin requests use transactional hold/reserve; finalization and voucher allocation are atomic or reconciled through an explicit saga with safe rollback.
- Game Expired/Archived is soft archive/public visibility change, not destructive deletion of evidence required for audit and My Games.
- Hazard geospatial indexes and route proximity checks must consider location accuracy/freshness to avoid false proximity.
- Saved prior-user trip plans preserve original creator and privacy; copy creates new owner draft.
- No sensitive secrets in analytics/export logs; admin datasets scoped to least privilege.

14. API / INTEGRATION GROUPS - FROM PROVIDED DOCX

இந்த 12 Groups, பயனர் வழங்கிய "Wayhunter_API_Integration_Groups_Tamil.docx" ஆவணத்திலிருந்து எடுக்கப்பட்டவை. கீழே உள்ள Links அதே ஆவணத்தில் குறிப்பிடப்பட்ட links. இவை API key/partner access கிடைத்துவிட்டதை உறுதிப்படுத்தவில்லை. Flutter native packages சிலவை வெளிப்புற Web API அல்ல.

01 Google Maps APIs

Document service: Android / iOS / Web Maps, Places Search, Routes, Geocoding, Navigation

Official/main link supplied: https://console.cloud.google.com/google/maps-apis/library

Wayhunter implementation use: Map & Places search in Flutter; platform keys restricted; routes backend/approved provider.

02 Weather APIs

Document service: Google Weather or Open-Meteo forecast; OpenWeather rain/cloud overlay

Official/main link supplied: https://console.cloud.google.com/apis/library/weather.googleapis.com

Wayhunter implementation use: Weather data/overlay with timestamps; validate availability and coverage.

03 Google API Keys

Document service: Maps/Places/Routes/Weather credentials and key restrictions

Official/main link supplied: https://console.cloud.google.com/apis/credentials

Wayhunter implementation use: Dedicated restricted keys per platform/API; never store server secrets in Flutter.

04 AI Trip Planner

Document service: Gemini API via Google AI Studio

Official/main link supplied: https://aistudio.google.com/app/apikey

Wayhunter implementation use: Server-mediated generation; editable plans, fallback and cost/quota monitoring.

05 Language & Translation

Document service: Flutter localization for UI; Cloud Translation for posts/captions

Official/main link supplied: https://console.cloud.google.com/apis/library/translate.googleapis.com

Wayhunter implementation use: UI translation built-in; dynamic text translation optional paid integration.

06 Email & OTP

Document service: Resend Email API

Official/main link supplied: https://resend.com/api-keys

Wayhunter implementation use: OTP/recovery provider; expiry/rate-limit enforced by Wayhunter backend.

07 Media Storage

Document service: Cloudinary images/video/live 360 evidence

Official/main link supplied: https://console.cloudinary.com/

Wayhunter implementation use: Signed upload authorization; private evidence vs public media.

08 Push Notifications

Document service: Firebase Cloud Messaging

Official/main link supplied: https://console.firebase.google.com/

Wayhunter implementation use: Follow, admin, trip, hazard, coins and voucher notifications.

09 Phone GPS & Camera

Document service: Flutter geolocator + camera packages

Official/main link supplied: https://pub.dev/

Wayhunter implementation use: Client device SDK/packages, not separate paid server APIs; enforce live capture.

10 Backend & Live Updates

Document service: ASP.NET Core API + SignalR

Official/main link supplied: https://learn.microsoft.com/en-us/aspnet/core/tutorials/first-web-api

Wayhunter implementation use: Wayhunter-owned APIs for state, chatty live trip notifications and admin status updates.

11 Ads & Sponsor Campaigns

Document service: Google AdMob + Admin sponsor manager

Official/main link supplied: https://admob.google.com/

Wayhunter implementation use: Banner or rewarded ads only per placement policy; compliance and consent.

12 Voucher Supplier

Document service: Thyaga corporate enquiry + Admin-managed voucher

Official/main link supplied: https://thyaga.lk/for-corporates

Wayhunter implementation use: Manual stock/contract first; no assumed API or automatic merchant redemption.

14.1 API Ownership and Security Policy

API-001 Flutter may hold only appropriately restricted public-client map keys; OTP, Gemini and Cloudinary signing credentials belong in backend/secure secrets store.

API-002 Admin portal calls Wayhunter ASP.NET Core APIs; it does not directly approve by editing SQL tables.

API-003 HTTP requests use HTTPS, access tokens and role verification; SignalR subscriptions verify trip/recipient authorization.

API-004 API errors never create fabricated routes, weather, verified rewards or real vouchers.

API-005 Rate limits, budget caps, key rotation, provider terms, quotas and monitoring defined before production.

API-006 Weather and hazard are distinct: weather forecast is provider data; hazards are user reports with own verification state.

API-007 Native contacts sharing uses OS chooser, not a Gmail or WhatsApp account connection; contacts discovery requires device permissions/consent.

API-008 Thyaga partner agreement and actual valid voucher inventory are prerequisites to advertising merchant redemption as live.

API-009 Provider prices and free tiers change; this supplied API document contains names and links, not confirmed production costs.

API-010 Restricted zones require curated authorized geofence data, not assumed as guaranteed Google Maps API output.

15. NON-FUNCTIONAL REQUIREMENTS, KPI & BUSINESS CONTROL

Nonfunctional Area	Release Requirement
Privacy	Explicit GPS sharing / contact consent; account public/private; evidence permissions.
Security	Password hashing, verified OTP, protected admin roles, device anti-abuse, anti-replay and secured API credentials.
Responsiveness	Support compact mobile layouts and Tamil/Sinhala/English length. Avoid overflow and unusable controls.
Reliability	Empty, offline, timeout, retry and provider failure state; no duplicated critical writes.
Performance	Lazy-load maps/media, paginate users/places/history, compress permitted media.
Financial Integrity	Append-only Coins ledger, reserve/approve/release, idempotent issue, audit and reconciliation.
Safety	No unsafe quests, restricted areas geofencing, hazard disclaimers, abuse reporting and TTL.
Backup/Recovery	Protected SQL backups, restore drill, private media retention, critical audit log protection.
Operations	Set measurable uptime/latency, review SLA, media max size and notification window before production.
Business	Reward-funded liability vs real revenue, voucher procurement, partner agreements, ad revenue and cloud budget tracked.

15.1 Management KPI Scorecard

KPI	Measurement
Growth	Verified registrations, MAU, 100 Coin welcome events, suspicious duplicate claims.
Friends	Sent/accepted follow requests, shared invite conversion, contacts consent rate.
Traveler	Saved / actually Started / Completed trips, historical reuse, trip reminders, hazard pop-ups.
Hunter	Pending/Approved/Rejected, median review time, approved 200 Coins cost.
Games	Created/approved/rejected/expired, completions approved, 100 Coins payout, anti-cheat rate.
Coins	Coins minted/reserved/debited/released, lifetime-earned levels, breakage and cost/liability.
Vouchers	Pending/approved/issued/used/expired, supplier stock and fulfillment delays.

KPI	Measurement
Revenue	Ads/sponsor invoices, partner costs, net margin, cost per eligible reward.
Platform	API errors, crash-free sessions, upload success, notification delivery rate.

16. TEST & ACCEPTANCE MATRIX

AT - Registration / ID

AT-REG-01 Same account cannot be issued two welcome bonuses
AT-REG-02 OTP expired/reused refused
AT-REG-03 Unique UserId remains same when display name edited
AT-REG-04 Non-admin login cannot access Admin portal
AT-REG-05 Logout revokes token
AT-REG-06 Account delete requires identity verification

AT - Friends

AT-FRI-01 Profile & Add Friends open same screen
AT-FRI-02 Trip draft preserved on return
AT-FRI-03 Followers vs Following counts accurate
AT-FRI-04 Private follow request requires approval
AT-FRI-05 Self-follow/duplicate blocked
AT-FRI-06 OS share sheet shows available Gmail/WhatsApp/SMS appropriately
AT-FRI-07 Contacts denied state works; discoverability/consent respected
AT-FRI-08 Accepted trip membership never silently enables GPS

AT - Traveler

AT-TRA-01 Trip submit alone does not rank/complete
AT-TRA-02 Planned reminder delivered without auto-start
AT-TRA-03 Start event timestamp exists
AT-TRA-04 Finish requires active start and ends live sharing
AT-TRA-05 Trip restart does not create duplicate completion
AT-TRA-06 1km Find on-demand only
AT-TRA-07 Copied suggested plan leaves original unchanged
AT-TRA-08 AI failure preserves input
AT-TRA-09 Verified/unverified hazard displayed accurately
AT-TRA-10 Active trip approaching hazard shows relevant pop-up
AT-TRA-11 Resolved/expired hazard no longer triggers
AT-TRA-12 Traveler leaderboard count ignores fake completion

AT - Hunter

AT-HUN-01 Gallery button not available for 360 or photo proof
AT-HUN-02 GPS district and accuracy checked
AT-HUN-03 200m distance alone does not auto reject
AT-HUN-04 Submit queues UserId+Name to Admin
AT-HUN-05 My Hunts shows Pending
AT-HUN-06 Admin Reject shows reason and awards 0
AT-HUN-07 Admin Approve publishes place and awards exactly 200 once
AT-HUN-08 Double approval retry no double payout
AT-HUN-09 Private evidence not exposed publicly

AT - Game

AT-GAM-01 Creator gallery clue upload allowed; player proof gallery blocked
AT-GAM-02 New game does not publish before admin approve
AT-GAM-03 Creation +100 only on approval
AT-GAM-04 Creation reject +0 and My Games status
AT-GAM-05 Creator cannot play own game
AT-GAM-06 Final play +100 only after admin approve
AT-GAM-07 Pending result no coins in game leaderboard
AT-GAM-08 One-month expiry disables public list
AT-GAM-09 Expired game persists in creator My Games
AT-GAM-10 Completed games remain in player view
AT-GAM-11 Retries do not duplicate coins
AT-GAM-12 Rejected suspicious proof no reward

AT - Leaderboards

AT-LEA-01 Global button disabled on Home/Profile/Friends/Wallet
AT-LEA-02 Game uses Game-earned coins only
AT-LEA-03 Hunter uses approved Hunter coins only
AT-LEA-04 Traveler uses valid completed trips only
AT-LEA-05 Ranking handles ties and private profiles

AT - Wallet

AT-WAL-01 No HP/RC/XP labels in new user/admin pages
AT-WAL-02 Level star bar progress 2000 lifetime earned
AT-WAL-03 Redemption does not reduce final balance before approval
AT-WAL-04 Reservation prevents two simultaneous overspends
AT-WAL-05 Approve finalizes debit once
AT-WAL-06 Reject releases reservation once
AT-WAL-07 My Vouchers shows pending/issued states
AT-WAL-08 Issued voucher shows true expiry and amount
AT-WAL-09 Voucher one-use validation
AT-WAL-10 Minimum 10000/greater policy confirmed before release
AT-WAL-11 Transaction history covers all stages

AT - Admin & API

AT-ADM-01 All 17 admin modules accessible per role
AT-ADM-02 Approval queues include immutable UserId+Name
AT-ADM-03 Old auto-approval disabled in Hunter and Games
AT-ADM-04 Ads active only in schedule; 3s rotation test
AT-ADM-05 Historical route eligibility uses completed ordered stops
AT-ADM-06 Sensitive edits require reason/audit
AT-ADM-07 Notifications status independent of push success
AT-ADM-08 Cloudinary private media protected
AT-ADM-09 Keys not exposed in Flutter/Admin logs
AT-ADM-10 Weather, AI, Maps and FCM failures have fallback

17. CONFLICT REGISTER & OWNER DECISIONS

Topic	Original Source Position	Updated Draft Decision / Open Issue
Hunter Approval	Admin spec auto-verifies valid Hunter submission	NEW: all new Hunts manual Admin review. USER DECISION.
Game Publication	Admin spec says valid game may auto-publish	NEW: all created Games manual Admin approval. USER DECISION.
Currencies	Old BRD HP/RC/XP and 100 HP=10 RC; 100 RC=Rs10	NEW: Coins only, 10k Coins=Rs100. USER DECISION.
Leaderboards	Old global bottom navigation + Hunter/Game ranks	NEW: module-specific 3 boards; global disabled on named pages. 4th bottom-nav slot TBD.
Game 1-month	Old BRD no fixed one-month life	NEW: expires in one month; creator record retained. Start clock/time semantics TBD.
Game winner	Admin PDF first verified winner; old BRD conditional one winner	100 coins per eligible admin-approved completion; single vs multiple winner conflict TBD.
App download bonus	100 Coins on download	Server first verified account +100 once recommended. Verify abuse and duplicates policy.
Redemption threshold	“10,000 coins ku mela” wording	Exact >10000 or ≥10000 TBD; draft uses k 10u ntil owner decides.
Cash vs voucher	Old initial scope excludes direct cash-out; new request says real money conversion	Cash payout method, partner/financial compliance, eligibility, fees, tax, KYC, contracts TBD. Do not promise cash-out yet.
Pending Coins	User says Coins should not decrease before approval	Visible balance unchanged pending, internally reserved to prevent double-spend. Confirm display labels.
Contacts discovery	New request My Contacts matched accounts	Requires privacy-safe consent, access rights and discoverability controls.
Hazard matching	New request pop-up near another report	Admin-configured radius, route match, age and throttle; community unverified label required.
Approval SLAs	None set	Set review time, timeouts, appeals, cancellation and escalation before launch.
Old screen tally	Original 70 views	New T-15 Traveler Leaderboard and shared friend/context may change count; confirm final inventory after UI audit.

17.1 Proposed Production Release Sequence

- Phase A: Identity and UserId, authentic login/OTP, single Coins Ledger, Admin RBAC and permissions.
- Phase B: Unified Friends + Trip draft/share/start/finish + privacy/location consent.
- Phase C: Hunter full live evidence + manual Admin approvals +200 Coins + My Hunts.
- Phase D: Game create/play + manual dual approvals + month expiry + My Games.
- Phase E: Three leaderboards and level progress + complete wallet request/hold/approve/issue.
- Phase F: Hazard pop-ups, reminders, contacts matching, integrations, Ads/Campaigns and analytics.
- Phase G: Partner contract and payment/voucher fulfillment readiness; QA, operations, supervisor signoff.

18. SOURCE TRACEABILITY & SIGN-OFF

Source	Used for
Wayhunter_Final_BRD.pdf; 18 pages; 05 Nov 2026	Prior mobile 70-view screen descriptions, scope, security, voucher workflow, primary UX, initial business rules.
WayHunter_Admin_Portal_Full_Module_Specification (1)(1).pdf; 20 pages	17 Admin modules, dashboards, safety, ads, place/route/game management and operations.

Source	Used for
Wayhunter_API_Integration_Groups_Tamil.docx; 1 page	12 named API/Integration groups and exact links as listed.
Product Owner updates in this conversation; 07 Nov 2026	Friends, Hunter/Game manual approval, game one-month, Leaderboards, Coins, User ID and Notifications new requirements.

Source precedence: newest explicit user-approved requirements override conflicting older text. Any item not expressly specified by the source is documented as a recommendation or an outstanding decision, not quietly presented as an approved feature.

Sign-off

Approver	Status / date
Product Owner / Team	Pending review of this revised v2.0 draft
Development Lead	Pending feasibility and architecture review
QA / Acceptance	Pending test coverage review
Supervisor / Formal Approver	Pending formal signature

END OF ORIGINAL v2.0 CONTENT - v2.1 CHANGE CONTROL ADDENDUM FOLLOWS | Tamil requirements draft | Do not infer that API accounts, commercial contracts or platform release are complete.

19. CHANGE CONTROL ADDENDUM v2.1 - VOUCHER ADMIN FLOW

Date: 10 November 2026 | Status: Working draft. This addendum clarifies voucher ingestion and records a financial state-machine conflict. It does not approve OCR/QR scanning, cash payout, or a debit timing policy.

19.1 Source precedence and traceability

- Reference A: Wayhunter Master BRD Revised Tamil v2.0, sections 09 (CO-010..CO-017), 11 (ADM-12/13), 13, 17.
- Reference B: Wayhunter 17 Admin Modules Functions & Working Process Tamil, page 13 (voucher intake), page 14 (redeem and fulfill), page 19 (pending decisions).
- Reference C: Angular Admin Architecture v3.0, Backend Architecture v3.0, ER Diagram v1.0 (proposed, not production schema).

19.2 Confirmed requirement versus proposal

- [SOURCE-SUPPORTED] Admin must manage supplier offer and voucher stock, manual entry, image/PDF upload/preview, source validation, unique voucher code and expiry before stock is offered. Supplier authenticity requires genuine agreement/manual verification.
- [PROPOSED ONLY] OCR extraction of voucher text and QR/barcode prefill. A scan result cannot certify validity and requires human review. Add only upon explicit approval.
- [UNRESOLVED VCH-DEC-01] BRD v2.0: final debit on Admin Approve; newer Admin Working Process: final debit at actual Fulfill/Issue. No automatic policy selection. Pending request reserves Coins while Posted Balance remains unchanged.
- [UNRESOLVED VCH-DEC-02] Implement OCR/QR/Barcode scanner now or later? Design remains gated until approved.
- [UNRESOLVED] >=10,000 vs >10,000 redemption threshold; cash payout channel; partner fulfillment agreement, all unchanged.

19.3 Proposed voucher states and data notes

- Source document: Uploaded → Validated → Admin Verified / Rejected. Private MediaAsset reference + uploader/verifier/time; no public direct source path.
- Inventory item: Draft/Unverified → Available → Reserved/Assigned → Issued → Used/Expired. Genuine unique code managed as hash/encrypted reference, not plain log text.
- Redemption request: Pending (held Coins) → UnderReview → Approved → Fulfilled, or Rejected/Cancelled. Debit timing MUST be settled before implementing Approved/Fulfilled transitions.
- SQL ERD: Keep existing 118 baseline tables. Suggested optional voucher-source and scan-result entities are extension proposals, not new mandatory signed-off schema.
- Audit: ActorUserId, target UserId, offer/request/inventory ID, old/new state, reason, timestamp and idempotency key for financial or sensitive actions.

19.4 Acceptance criteria / non-negotiable safety

- Image/PDF upload rejects unsupported types/over-limit size; protected file preview; no voucher code plaintext exposed to unauthorized users.
- OCR/scan candidate never directly creates issued/usable voucher. Admin verifies code uniqueness, expiry, amount, supplier and offer terms.
- On Pending redemption: Posted balance unchanged, Reserved increases, Spendable decreases. Reject/Cancel releases once, no double credit.
- Approved-but-unissued voucher is not usable. One inventory item cannot be assigned to two users. Issued QR or code becomes visible only after actual assignment.
- Both final debit policies must be testable in sandbox as alternatives; production financial finalization blocked until VCH-DEC-01 is explicitly resolved.

Change note: Other 17 Admin Modules and Coins award rates remain unchanged. Document remains Draft for Approval.